Thrust-to-Weight Ratio & Wing Loading

1 The two key parameters

T/W and W/S affect aircraft performance more than any other parameters and must be chosen *together*.

- **Low** W/S (large wing): room for gear and fuel, but more wetted area and drag.
- **High** W/S (small wing): less drag, but gear and fuel may not fit.
- **Short take-off:** large wing and small engine, or small wing and large engine.

Procedure: estimate T/W first, find W/S from the critical requirements (often approach stall speed), then recheck T/W.

2 Thrust-to-weight ratio

Definition: sea level, zero velocity (static), standard day, design take-off weight, maximum throttle. A value found at any other condition must be converted back.

Higher T/W: quicker acceleration, faster climb, higher top speed and turn rate, but more fuel and heavier engines. T/W varies in flight (W falls, $T = T(h, V)$).

Propeller aircraft use power loading W/hp (most aircraft 10–15 lb/hp; aerobatic about 6):

$$\frac{T}{W} = \frac{\eta_p P}{VW} = \frac{550\,\eta_p}{V}\,\frac{hp}{W} \quad \text{(fps units)}$$

Jet aircraft	T/W	Propeller aircraft	W/hp
Jet trainer	0.4	Powered sailplane	25
Jet fighter (dogfighter)	0.9	Homebuilt	12
Jet fighter (other)	0.6	GA single engine	14
Military cargo/bomber	0.25	GA twin engine	6
Jet transport	0.25	Agricultural	11
		Twin turboprop	5
		Flying boat	10

Curve fits: $T/W_0 = a\,M_{max}^{C}$ (jets); $hp/W_0 = a\,V_{max}^{C}$ (V_{max} in mph, hp/lb).

Jet	a	C	Propeller	a	C
Trainer	0.488	0.728	Sailplane	0.043	0
Fighter (dogf.)	0.648	0.594	Homebuilt, metal	0.005	0.57
Fighter (other)	0.514	0.141	Homebuilt, comp.	0.004	0.57
Cargo/bomber	0.244	0.341	GA single	0.024	0.22
Transport	0.267	0.363	GA twin	0.034	0.32
			Agricultural	0.008	0.50
			Twin turboprop	0.012	0.50
			Flying boat	0.029	0.23

3 Thrust matching (cruise-efficient aircraft)

In cruise $L = W$ and $T = D$, so $(T/W)_{cruise} = 1/(L/D)_{cruise}$, with $(L/D)_{cruise} = 0.866\,(L/D)_{max}$ (jet) or $(L/D)_{max}$ (prop). Convert to take-off:

$$\left(\frac{T}{W}\right)_{TO} = \frac{1}{(L/D)_{cruise}}\,\frac{W_{cruise}}{W_{TO}}\,\frac{T_{TO}}{T_{cruise}}$$

$$\left(\frac{hp}{W}\right)_{TO} = \frac{V_{cr}}{550\,\eta_p}\,\frac{1}{(L/D)_{cr}}\,\frac{W_{cr}}{W_{TO}}\,\frac{hp_{TO}}{hp_{cr}}$$

- $W_{cruise}/W_{TO} \approx 0.970 \times 0.985 = 0.956$ (start of cruise).
- **Thrust lapse** (Fig. 5.1): high-bypass turbofan, cruise thrust 20–25% of take-off; low-bypass afterburning, 40–70%.
- **Piston:** power falls with density; unsupercharged at 10,000 ft gives about 73% of sea-level power; turbo/supercharging holds it to a limit (Fig. 5.2). Cruise at about 75% power.
- **Turboprop:** equivalent shaft hp; cruise 60–80% of take-off.

Example: $(L/D) = 17$, lapse 0.225: $T/W = (1/17)(0.956)/0.225 \approx 0.25$.

4 Climb often sets T/W

$$\left(\frac{T}{W}\right)_{climb} = \frac{1}{(L/D)} + \underbrace{\frac{V_{vertical}}{V}}_{G,\ \text{climb gradient}}$$

Example: twin, one engine out, $G \geq 2.4\%$ (FAR 25), $L/D \approx 10$: the remaining engine needs $0.124\,W$, so installed $T/W \approx 0.25$ while cruise needs only about 0.06. First pass: take the higher of the statistical and thrust-matching values.

5 Wing loading basics

$W/S = W_{TO}/S_{ref}$ (whole reference wing). It is the average pressure difference across the wing. A larger wing improves stall, climb, take-off, landing and turning, but adds drag and empty weight, which the sizing equation amplifies.

Aircraft	lb/ft^2	kg/m^2
Sailplane	6	30
Homebuilt	11	54
GA single engine	17	83
GA twin engine	26	127
Twin turboprop	40	195
Jet trainer	50	244
Jet fighter	70	342
Jet transport/bomber	120	586

Faster aircraft have higher W/S. $\times 4.88$ for kg/m^2, $\times 47.9$ for N/m^2.

Rule: convert every requirement's W/S to take-off conditions and select the **lowest**. If one requirement forces an unreasonable value, change the design instead (better high-lift system, more thrust).

6 Stall speed

$$\frac{W}{S} = \tfrac{1}{2}\,\rho\,V_{stall}^2\,C_{L_{max}}$$

ρ: 0.00238 slug/ft^3 at sea level, or 0.00189 for 5,000 ft hot day. FAR 23: $V_{stall} \leq 61$ kt; trainers about 50 kt. $V_{approach} \geq 1.3\,V_{stall}$ (civil), 1.2 (military), 1.15 (carrier).

Configuration	$C_{L_{max}}$
Plain wing, no flaps	1.2–1.5
Inboard flaps	1.6–2.0
Transport, flaps and slats	2.4
STOL	3.0
Large flaps in propwash/jetwash	up to 5.0

Take-off $C_{L_{max}} \approx 80\%$ of landing value. For $A > 5$, wing $C_{L_{max}} \approx 0.9\,C_{l_{max}}$. Partial-span flaps: $C_{L_{max}} \approx 0.9\left[(C_{l_{max}})_{fl}\,\frac{S_{fl}}{S_{ref}} + (C_l)_{unfl}\,\frac{S_{unfl}}{S_{ref}}\right]$.

7 Take-off distance

Lift-off at $1.1\,V_{stall}$, so $C_{L_{TO}} = C_{L_{max}}/1.21$. Obstacle: 50 ft (military, small civil), 35 ft (commercial). Balanced field length and decision speed V_1 cover engine failure.

$$TOP_{jet} = \frac{W/S}{\sigma\,C_{L_{TO}}\,(T/W)}, \quad TOP_{prop} = \frac{W/S}{\sigma\,C_{L_{TO}}\,(hp/W)}$$

Read TOP for the required distance from Fig. 5.4, then

$$\frac{W}{S} = (TOP)\,\sigma\,C_{L_{TO}}\,\frac{T}{W} \quad \text{or} \quad (TOP)\,\sigma\,C_{L_{TO}}\,\frac{hp}{W}$$

More T/W allows a higher W/S; hot or high fields (low σ) need a lower W/S.

Catapult: $\left(\frac{W}{S}\right)_{TO} = \frac{1}{2}\rho(V_{end} + V_{wod})^2\,\frac{(C_{L_{max}})_{TO}}{1.21}$, wind over deck 20–40 kt.

8 Landing distance

Distance $\propto V^2_{touchdown}$: double the speed, four times the runway. First guess: $S_{landing} \approx 0.3\, V^2_{approach}$ (ft, kt).

$$S_{landing} = 80\,\frac{W/S}{\sigma\, C_{L_{max}}} + S_a$$

S_a: 1000 ft (airliner, 3°), 600 ft (GA), 450 ft (STOL, 7°). Reversers: ground term ×0.66. FAR 25: total ×1.67. Then

$$\left(\frac{W}{S}\right)_{land} = \frac{\sigma C_{L_{max}}(S_{landing} - S_a)}{80}$$

$$\left(\frac{W}{S}\right)_{TO} = \frac{(W/S)_{land}}{W_{land}/W_{TO}}$$

$W_{land}/W_{TO} \approx 1.0$ (props, jet trainers), 0.85 (most jets).
Arrested landing: engagement speed (Fig. 5.6) ÷1.15 gives V_{stall}, then the stall equation.

9 Cruise and loiter

$$D = qSC_{D_0} + qS\,\frac{C_L^2}{\pi Ae}, \qquad C_L = \frac{W/S}{q}$$

First estimates: $C_{D_0} = 0.015$ (clean jet), 0.02 (clean prop), 0.03 (prop, fixed gear); $e = 0.6$ (fighter), 0.8 (others).

Mission	Aircraft	Optimum	W/S
Range	Prop	parasite = induced	$q\sqrt{\pi AeC_{D_0}}$
Range	Jet	parasite = 3× induced	$q\sqrt{\pi AeC_{D_0}/3}$
Loiter	Jet	parasite = induced	$q\sqrt{\pi AeC_{D_0}}$
Loiter	Prop	induced = 3× parasite	$q\sqrt{3\pi AeC_{D_0}}$

Jet best range: 1.32× the best-L/D speed for 13% less L/D. Fuel burn lowers W/S, so the aircraft cruise-climbs (or stair-steps). Divide by $W_{avg}/W_{TO} \approx 0.85$. These optimise only part of the mission: ignore them if far below Table 5.5.

10 Turning

Level turn: $L\cos\varphi = W$, load factor $n = L/W$.

$$\dot\psi = \frac{g\sqrt{n^2-1}}{V}, \qquad R = \frac{V^2}{g\sqrt{n^2-1}}, \qquad n = \frac{q\,C_L}{W/S}$$

Instantaneous turn (lift and structure limited):

$$n = \sqrt{\left(\frac{\dot\psi V}{g}\right)^2 + 1}, \qquad \frac{W}{S} = \frac{q\,C_{L_{max}}}{n}$$

with combat $C_{L_{max}} \approx 0.6$–0.8 (TE flap) or 1.0–1.5 (LE + TE flaps). Combat/take-off weight ≈ 0.85. **Corner speed**: where $C_{L_{max}}$ gives the limit load factor (7.33 g; newer 8–9 g); the best turn rate.
Sustained turn ($T = D$): $n = (T/W)(L/D)$,

$$\frac{T}{W} = \frac{qC_{D_0}}{W/S} + \frac{W}{S}\,\frac{n^2}{q\pi Ae}, \qquad \left(\frac{W}{S}\right)_{min\,T/W} = \frac{q}{n}\sqrt{\pi AeC_{D_0}}$$

$$\frac{W}{S} = \frac{T/W \pm \sqrt{(T/W)^2 - 4n^2C_{D_0}/(\pi Ae)}}{2n^2/(q\pi Ae)}$$

Use the + root. No solution unless $T/W \geq 2n\sqrt{C_{D_0}/(\pi Ae)}$. Results are very sensitive to e.

11 Climb, glide and ceiling

Climb gradient $G = (R/C)/V = (T-D)/W$.

$$\frac{W}{S} = \frac{(T/W - G) \pm \sqrt{(T/W-G)^2 - 4C_{D_0}/(\pi Ae)}}{2/(q\pi Ae)}$$

$$\frac{T}{W} \geq G + 2\sqrt{\frac{C_{D_0}}{\pi Ae}}$$

Configuration	ΔC_{D_0}	e
Take-off flaps	+0.02	−5%
Landing flaps	+0.07	−10%
Gear down	+0.02	–

One engine out: a twin keeps 1/2, a three-engine aircraft 2/3, of T/W. Glide: $T/W = 0$, $G < 0$. Ceiling: $G = 0$ (or a small G for 100 ft/min). Very high altitude: $W/S = qC_L$, with $C_L \approx 0.5$ (1.0–1.4 for high-lift aerofoils).

12 Selecting the design point

1. Estimate T/W (statistics, thrust matching, climb).
2. Find the W/S limit from each requirement.
3. Convert each to take-off conditions.
4. Select the **lowest** W/S.
5. If one requirement drives it, change the design, not just the wing.
6. Recheck T/W with the chosen W/S; compare with real aircraft.

A matching (constraint) diagram plots every requirement as T/W against W/S; the design point is the lowest T/W in the allowed region.